

Write legibly showing all the steps VERY clearly. (Total points = 100).

(1) Define the following: Let Ω be a non-empty set.

- (a) $\mathcal{F}$, a collection of subsets of Ω , is called a σ -algebra if –
- (b) A set function P , defined on a σ -algebra $\mathcal{F}$ of Ω , is a *probability* measure if –
- (c) A function X defined on a prob. space $(\Omega, \mathcal{F}, P)$ is a random variable (r.v) if –
- Let $\{X_n\}$, X be random variables defined on some $(\Omega, \mathcal{F}, P)$.
- (d) X_n converges to X “almost surely (a.s)” or “with probability 1 (w.p. 1)” if –
- (e) X_n converges to X in probability (w.r.t P) if –
- (f) X_n converges to X in L^p , for $0 < p < \infty$ if –
- (g) if each $X_n \in L^1(\Omega, \mathcal{F}, P)$, then $\{X_n\}$ is called uniformly integrable (UI) if –
- Let X, Y be two r.v's defined on $(\Omega, \mathcal{F}, P)$, with c.d.f F_X, F_Y resp.ly, then
- (h) X is a *Exponential*(1) random variable if –
- (i) X is called a *discrete* random variable if –
- (j) the two random variables X and Y are called independent if –

(4×10 = 40 points)

(2) Prove the following:

- (a) Let $(\Omega, \mathcal{F})$ be a measurable space and μ_0, μ_1, ν_0 and ν_1 be σ -finite measures. If $\mu_0 \ll \mu_1$, $\nu_0 \ll \nu_1$ and $\mu_1 \perp \nu_1$, then show that $\mu_0 \perp \nu_0$.
- (b) Let X, Y be two independent r.v. defined on $(\Omega, \mathcal{F}, P)$ with $P_X(A) = \sum_{n=0}^{\infty} p_n \mathbf{I}_A(n)$ and $P_Y(A) = \sum_{n=0}^{\infty} q_n \mathbf{I}_A(n)$ for $A \in \mathcal{B}(\mathbb{R})$. Here $\tilde{p} = \{p_n\}_{n \geq 1}$, $\tilde{q} = \{q_n\}_{n \geq 1}$ are two nonnegative sequences with $\sum_{n=0}^{\infty} p_i = \sum_{n=0}^{\infty} q_i = 1$.

Then show $P_{(X+Y)}(A) = \sum_{n=0}^{\infty} (\tilde{p} * \tilde{q})(n) \mathbf{I}_A(n)$ for $A \in \mathcal{B}(\mathbb{R})$.

(10+10=20 points)

(3) Let $\Omega = (0, 1)$, $\mathcal{F} = \mathcal{B}((0, 1))$ and P be the Lebesgue measure. Let $X(\omega) = -\log \omega$ and $Z = h(X)$, where $h(x) = x^2$. Let $Y(\omega) = \mathbf{I}_{\{(\frac{1}{3}, 1)\}}(\omega)$ and be independent of X .

- (a) Find P_X, P_Y and P_Z on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$
- (b) Argue the existence of a sequence $\{Z_n\}$ of i.i.d. random variables each having the same distribution as Z in (a). State the result(s) clearly in justifying this statement.
- (c) For $\{Z_n\}$ as in (b), define for each $n \geq 1$, $T_n(\omega) = (\frac{1}{n} \sum_{i=1}^n Z_i(\omega))$. Argue that T_n is measurable and compute (give a numerical answer) for each $n \geq 1$

$$\int_{\Omega} T_n(\omega) dP(\omega).$$

- (d) What is the distribution of $X + Y$? Is it absolutely continuous w.r.t m ? If so, express the RN derivative function $\left(\frac{P_{(X+Y)}}{dm}\right)(x)$ for each $x \in \mathbb{R}$.

(10+10+10+10=40 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A)$, $A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1$, $n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A)$, $A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$, $a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$